



**Module 1: Optimization Modeling; Excel Implementation using Solver
Lab Problem Set**

Required Problems: 1-3

Additional optional problems: 4-10

Problems labelled with an asterisk (*) are more challenging

Problem 1

The Molokai Nut Company (MNC) makes four different products from macadamia nuts grown in the Hawaiian Islands: chocolate-coated whole nuts (Whole), chocolate-coated nut clusters (Cluster), chocolate-coated nut crunch bars (Crunch), and plain roasted nuts (Roasted). The company is barely able to keep up with the increasing demand for these products. However, increasing raw material prices and foreign competition are forcing MNC to watch its margins to ensure that it is operating in the most efficient manner possible. To meet marketing demands for the coming week, MNC needs to produce at least 1,000 pounds of the Whole product, between 400 and 500 pounds of the Cluster product, no more than 150 pounds of the Crunch product, and no more than 200 pounds of Roasted product. Each pound of the Whole, Cluster, Crunch, and Roasted product contains, respectively, 60%, 40%, 20%, and 100% macadamia nuts, with the remaining weight made up of chocolate coating. The company has 1,100 pounds of nuts and 800 pounds of chocolate available for use in the next week. The various products are made using four different machines that hull the nuts, roast the nuts, coat the nuts in chocolate (if needed), and package the products. The following table summarizes the time required by each product on each machine. Each machine has 60 hours of time available in the coming week.

Machine	Minutes required per pound			
	Whole	Cluster	Crunch	Roasted
Hulling	1.00	1.00	1.00	1.00
Roasting	2.00	1.50	1.00	1.75
Coating	1.00	0.70	0.20	0.00
Packaging	2.50	1.60	1.25	1.00

The selling price and variable cost associated with each pound of product is summarized below:

	Whole	Cluster	Crunch	Roasted
Selling Price	\$5.00	\$4.00	\$3.20	\$4.50
Variable Cost	\$3.15	\$2.60	\$2.16	\$3.10

- Formulate an LP model for this problem.
- Create a spreadsheet model for this problem and solve it using Solver.
- What is the optimal solution?

Problem 2

Maintenance at a major theme park in central Florida is an ongoing process that occurs 24 hours a day. Because it is a long drive from most residential areas to the park, employees do not like to work shifts of fewer than eight hours. These 8-hour shifts start every four hours throughout the day. The number of maintenance workers needed at different times throughout the day varies. The following table summarizes the minimum number of employees needed in each 4-hour time period.

Time Period	Minimum Employees Needed
12 a.m. to 4 a.m.	90
4 a.m. to 8 a.m.	215
8 a.m. to 12 p.m.	250
12 p.m. to 4 p.m.	165
4 p.m. to 8 p.m.	300
8 p.m. to 12 a.m.	125

The maintenance supervisor wants to determine the minimum number of employees to schedule that meets the minimum staffing requirements.

- Formulate an LP model for this problem.
- Create a spreadsheet model for this problem and solve it using Solver.
- What is the optimal solution?

Problem 3

Holiday Fruit Company buys oranges and processes them into gift fruit baskets and fresh juice. The company grades the fruit it buys on a scale from 1 (lowest quality) to 5 (highest quality). The following table summarizes Holiday's current inventory of fruit.

Grade	Supply (1000s of lbs)
1	90
2	225
3	300
4	100
5	75

Each pound of oranges devoted to fruit baskets results in a marginal profit of \$2.50, whereas each pound devoted to fresh juice results in a marginal profit of \$1.75. Holiday wants the fruit in its baskets to have an average quality grade of at least 3.75 and its fresh juice to have a average quality grade of at least 2.50

- Formulate an optimization model for this problem.
- Implement your model in a spreadsheet and solve it.
- What is the optimal solution?

Problem 4

A bank has \$650,000 in assets to allocate among investments in bonds, home mortgages, car loans, and personal loans. Bonds are expected to produce a return of 10%, mortgages 8.5%, car loans 9.5%, and personal loans 12.5%. To make sure the portfolio is not too risky, the bank wants to restrict personal loans to no more than the 25% of the total portfolio. The bank also wants to ensure that more money is invested in mortgages than in personal loans. It also wants to invest more in bonds than personal loans.

- a. Formulate an LP model for this problem with the objective of maximizing the expected return on the portfolio.
- b. Implement your model in a spreadsheet and solve it.
- c. What is the optimal solution?

Problem 5

Valu-Com Electronics manufactures five different models of telecommunications interface cards for personal and laptop computers. As summarized in the following table, each of these devices requires differing amounts of printed circuit (PC) board, resistors, memory chips, and assembly.

	Per Unit Requirements				
	HyperLink	FastLink	SpeedLink	MicroLink	EtherLink
PC Board (sq. in.)	20	15	10	8	5
Resistors	28	24	18	12	16
Memory Chips	8	8	4	4	6
Assembly Labor	0.75	0.6	0.5	0.65	1

The unit wholesale price and manufacturing cost for each model are as follows.

	Per Unit Revenues and Costs				
	HyperLink	FastLink	SpeedLink	MicroLink	EtherLink
Wholesale Price	\$189	\$149	\$129	\$169	\$139
Manufacturing Cost	\$136	\$101	\$ 96	\$137	\$101

In their next production period, Valu-Com has 80,000 square inches of PC board, 100,000 resistors, 30,000 memory chips, and 5,000 hours of assembly time available. The company can sell all the product it can manufacture, but the marketing department wants to be sure that it produces at least 500 units of each product and at least twice as many FastLink cards as HyperLink cards while maximizing profit.

- Formulate an LP model for this problem.
- Create a spreadsheet model for this problem and solve it using Solver.
- What is the optimal solution?

Problem 6

Riverside Oil Company in eastern Kentucky produces regular and supreme gasoline. Each barrel of regular sells for \$21 and must have an octane rating of at least 90. Each barrel of supreme sells for \$25 and must have an octane rating of at least 97. Each of these types of gasoline are manufactured by mixing different quantities of the following three inputs:

Input	Cost per Barrel	Octane Rating	Barrels Available (in 1000s)
1	\$17.25	100	150
2	\$15.75	87	350
3	\$17.75	110	300

Riverside has orders for 300,000 barrels of regular and 450,000 barrels of supreme. How should the company allocate the available inputs to the production of regular and supreme gasoline if it wants to maximize profits?

- Formulate an LP model for this problem.
- Create a spreadsheet model for this problem and solve it using Solver.
- What is the optimal solution?

Problem 7

The Weedwacker Company manufactures two types of lawn trimmers: an electric model and a gas model. The company has contracted to supply a national discount retail chain with a total of 30,000 electric trimmers and 15,000 gas trimmers. However, Weedwacker's production capability is limited in three departments: production, assembly, and packaging. The following table summarizes the hours of processing time available and the processing time required by each department, for both types of trimmers:

	Hours Required Per Trimmer		Hours Available
	Electric	Gas	
Production	0.20	0.40	10,000
Assembly	0.30	0.50	15,000
Packaging	0.10	0.10	5,000

The company makes its electric trimmer in-house for \$55 and its gas trimmer for \$85. Alternatively, it can buy electric and gas trimmers from another source for \$67 and \$95, respectively. How many gas and electric trimmers should Weedwacker make and how many should it buy from its competitor to fulfill its contract in the least costly manner?

- Formulate an LP model for this problem.
- Create a spreadsheet model for this problem and solve it using Solver.
- What is the optimal solution?

Problem 8*

The Springfield City Police Department employs 30 police officers. Each officer works five days per week. The crime rate fluctuates with the day of the week, so the number of police officers required each day depends on the day of the week, as follows: Saturday, 28; Sunday, 18; Monday, 18; Tuesday, 24; Wednesday, 25; Thursday, 16; Friday, 21. The police department wants to schedule police officers to minimize the number of police officers whose days off are not consecutive. Formulate an optimization model to accomplish this goal.

Problem 9*

The Swift & Kelce music band will start touring soon. Their manager has arranged for the band to give a concert at the Enterprise Center in St. Louis, an indoor concert venue. The floor area of the venue is 100,000 square feet and can be arranged to accommodate 3 types of tickets: Premium, Standard and Standing Room. Each type of ticket takes up a different amount of floor space: premium tickets 8 square feet, standard tickets 6 square feet and standing room tickets 3 square feet. The manager estimates that the premium tickets can sell for \$500, the standard tickets for \$250 and the standing room tickets for \$100. Assume the concert will sell out.

Due to existing fire code regulations, a maximum of 20,000 tickets can be sold. Additionally there is a combined upper limit of 8,000 total premium and standard tickets that can be sold to ensure a large enough standing room only crowd in front of the stage. Finally, according to Center regulations there must be at least 3 standard seats for every 1 premium seat.

- a. Write down an optimization model to maximize revenue subject to the above constraints. Please be explicit about your decision variables and constraints.
- b. Find the optimal solution using Excel's Solver. What is the optimized concert revenue?
- c. The band manager has approached the fire marshall to negotiate the capacity of 20,000. The fire marshall responded that he would consider increasing the capacity if the band manager invested in additional safety measures. How much should the band manager be willing to pay for additional safety measures to allow 400 additional tickets to be sold?

Problem 10*

In planning an in-person final exam for one of our courses, we needed to determine how many students are allocated to each exam classroom. In true Analytics spirit, let's view this as an optimization problem.

Suppose we have 6 sections of the course with the following number of students enrolled in each section:

Section	A	B	C	D	E	F
Enrollment	63	62	63	61	62	60

We have 6 exam classrooms with the following seat capacities:

Classroom	1	2	3	4	5	6
Capacity	86	86	76	128	54	70

We can assign any student from any section to take the exam in any classroom.

In this problem we will assume that the objective function is to minimize the density of the highest density classroom. The density of a classroom is defined to be the number of students allocated to that classroom divided by the classroom's capacity.

We require that each classroom holds at most 70 students. This would allow the TA assigned to the classroom to distribute and collect exam papers quickly at the start and end of the exam.

- Formulate the problem as a linear optimization problem. Define your decision variables clearly and write down the objective function and constraints. (Just provide the formulation, you don't need to solve this problem numerically.)
Hint: Include constraints of the form: "density of classroom $i \leq z$ " where z is a decision variable representing an upper limit on classroom density, and "density of classroom i " is expressed in terms of your remaining decision variables. Then write the objective function in terms of z .
- Ignoring the constraint that limits the number of students allocated to each classroom to 70, what is the optimal objective function value of the formulation of part (a)? That is, what is the smallest possible density of the highest density classroom. (You don't need Solver!)